

# Rishi More

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## Summary

**ML systems engineer** specializing in efficient and reliable LLM inference. **ICML 2026 co-first author** with hands-on experience in CUDA/Triton kernels, Nsight profiling, INT4 quantization, vLLM serving, and distribution-free risk control.

## Experience

### JHU Data Science and AI Institute (DSAI)

May 2025 – June 2026

*Graduate Research Assistant, LLM Inference Optimization*

*Baltimore, Maryland*

- Co-first authored *Conformal Thinking (ICML 2026 main track)*, introducing finite-sample calibrated early stopping for reasoning LLMs while controlling a target error rate.
- Built a vLLM-based calibration and evaluation pipeline across **4 reasoning models**, **4 benchmarks**, and **2.7K+ problems**, maintaining target error rates under dataset and response-length shift with finite-sample UCB calibration.
- Designed dual-threshold and ensembled halt policies that reduced reasoning tokens by **12% at matched accuracy**, with reusable tooling for calibration, evaluation, and ablation studies.

### JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

*Graduate Researcher, Retrieval Systems*

*Baltimore, Maryland*

- Built temporal RAG over **10K+ timestamped documents** using revision-aware chunk tracking, time-decay scoring, and metadata-scoped indexing to prioritize current evidence.
- Improved retrieval precision by **23%** over dense-only retrieval and reduced stale or hallucination-prone contexts by **31%** through recency weighting and reference-free consistency scoring.

### Mastek

November 2022 – July 2023

*Machine Learning Engineer Intern*

*Mumbai, India*

- Deployed an on-premises edge voice-authentication system with **<100 ms p99 latency** and **96.6% accuracy**, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment.
- Built a TensorFlow pipeline over **1M+ voice samples**; spectral-gating denoising, augmentation, and robust feature extraction improved authentication reliability by **20%** in high-noise conditions.

## Projects

### Conformal Serve: Risk-Controlled Adaptive LLM Serving | vLLM, FastAPI, Docker Compose, Prometheus

- Built an engine-agnostic adaptive-stopping core shared by deterministic mock-engine CI and vLLM's AsyncLLMEngine; achieved **100% policy branch coverage** with `mypy --strict` and exposed client-set risk  $\alpha$  through an OpenAI-compatible API.
- On Qwen2.5-1.5B-Instruct and an 8GB RTX 4070, reduced reasoning tokens **64.7% at  $\alpha = 0.10$**  and **71.4% at  $\alpha = 0.30$**  while maintaining realized error  $\leq \alpha$  on small held-out arithmetic sets.
- Calibrated per- $\alpha$  thresholds offline by distribution-free risk control into immutable, content-addressed artifacts; shipped Docker Compose with Prometheus/Grafana and a one-command benchmark reporting **95%** Clopper–Pearson error bounds.

### INT4 KV-Cache Quantization & Fused Triton Attention for Long-Context Decode | Triton, CUDA, PyTorch

- Built a correctness-gated fused Triton flash-decode dequant-attention kernel, cutting 32K-context attention latency **13.7 $\times$  (7.54 to 0.55 ms/layer)** versus PyTorch FP16 GQA SDPA while maintaining **0.999 cosine similarity** to the dequantized oracle.
- Reduced KV storage from **28.7 KB to 8.3 KB per token (3.47 $\times$ )**, raising estimated 8GB KV capacity from **~154K to ~534K tokens**; retained **100% passkey retrieval through 32K** with only **+1.0% WikiText perplexity**.
- Diagnosed per-token-K failure as outlier-channel collapse (attention cosine **0.08 vs. 0.98**) and built an RCPS controller certifying **~50% cache eviction at 19% realized loss** under a **30% risk budget**.

### Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | CUDA C++, Nsight, vLLM

- Profiled Qwen2.5-1.5B batch-1 decode with Nsight; attributed **~81% of GPU-kernel time** to weight GEMVs and measured dominant MLP projections at **~247 GB/s** against a **~250 GB/s** DRAM ceiling.
- Built a correctness-gated fused CUDA INT4 group-dequantization + GEMV kernel reaching **~223 GB/s**; accelerated batch-1 MLP projections by **2.38 $\times$**  versus FP16 GEMM and mapped the crossover to batch 2 as weight reuse made GEMM faster.

## Education

### Johns Hopkins University

May 2026

*M.S.E. in Computer Science*

*GPA: 3.9/4.0*

### University of Mumbai

July 2024

*B.Tech. in Computer Science, Honors in AI/ML*

*CGPA: 9.85/10*

## Technical Skills

**Languages:** Python, C++, CUDA C++, C, SQL, Bash

**ML/LLM Systems:** PyTorch, Triton, vLLM, Hugging Face Transformers, TensorFlow, LoRA, DPO, RAG

**Performance & Infrastructure:** Nsight Compute/Systems, Linux, Git, Docker, GitHub Actions, AWS

**Methods:** Quantization, kernel fusion, roofline analysis, LLM evaluation, conformal risk control, calibration